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Similarity Solution for a Spherical Shock Wave

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The point-source, spherical shock wave moving into a constant density γ-law gas is considered in the limit of infinite shock strength from the point of view of the Richtmyer-von Neumann viscosity technique. A similarity solution of this problem is shown to exist and is obtained for various boundary conditions with $\gamma=1.4$. The solutions are obtained analytically in that part of the flow field not involving viscosity, and numerically in the other parts of the flow field. It is found that whereas all discontinuities of the physical parameters are removed by the viscosity there remain discontinuities in the slopes of these parameters at the shock front. It is indicated, moreover, that the complete flow field depends upon the form and magnitude of the viscosity.